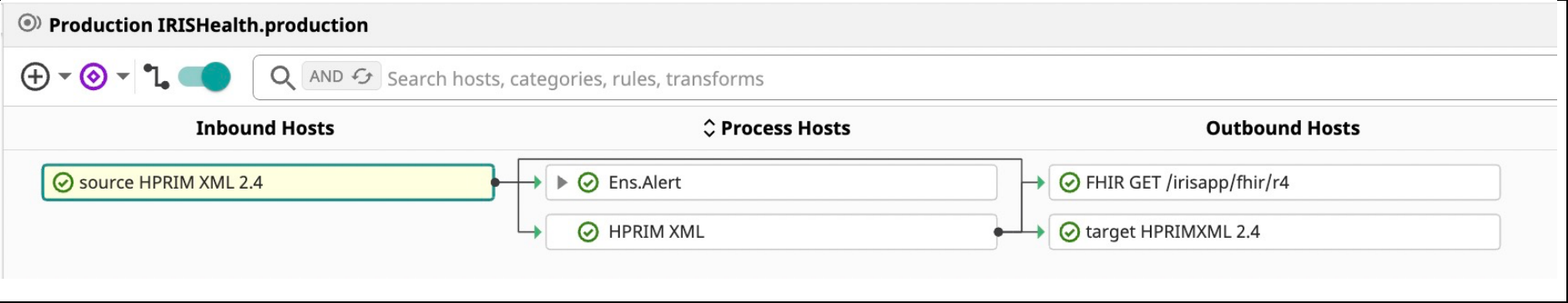


Production
-
Flux HPRIM XML en source et cible avec processus requêtant un serveur FHIR pour enrichir des données manquantes ou erronées.



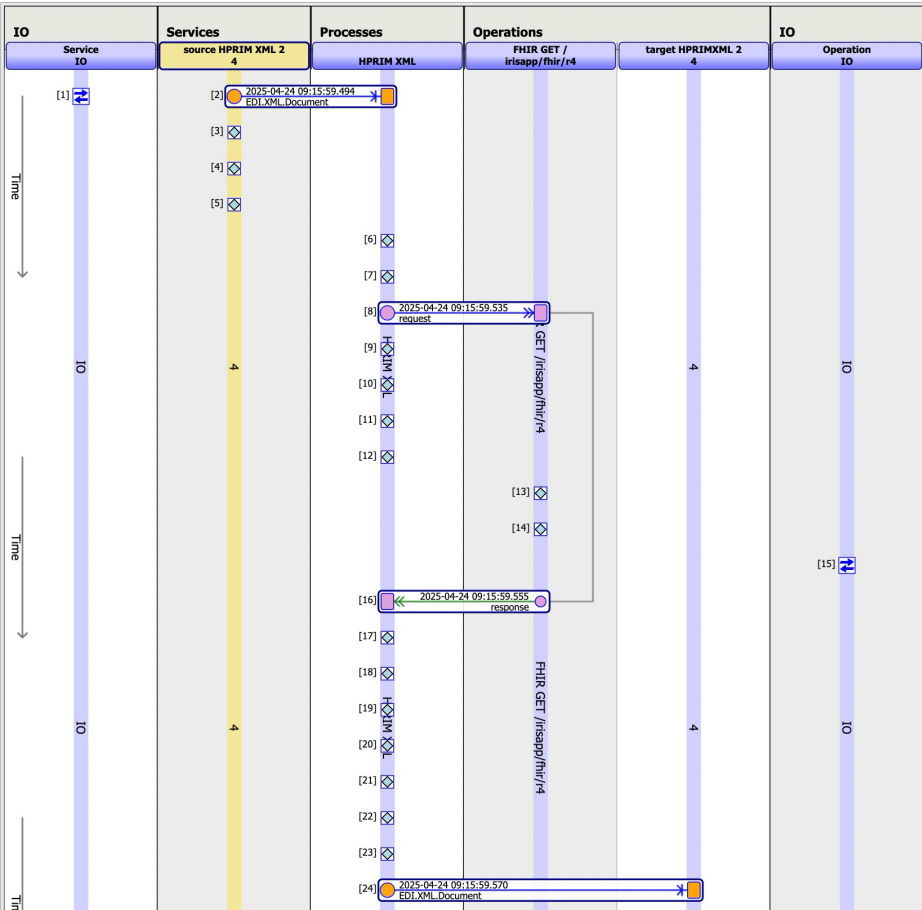
Visual Trace

- Étape Service
Capture document
HPRIM XML

- Document HPRIM XML
initial sans date de
naissance

VISUAL TRACE

Session ID: 24108 Legend Printable Version Go to items 1 - 34 Items per page 200 Show events Show internal items Apply Filter None Previous Page Next Page Previous Session Next Session



```
<?xml version="1.0" ?>
<!-- type: EnsLib.EDI.XML.Document id: 64 -->
<xsd_2:evenementsServeurActes xmlns:xsd_3="http://www.hprim.org/inseeXML"
xmlns:xsd_2="http://www.hprim.org/hprimXML">
  <xsd_2:enteteMessage>
    <xsd_2:dateHeureProduction>2025-04-24 01:22:44</xsd_2:dateHeureProduction>
    <xsd_2:emetteur>
      <xsd_2:agent>GigaComp Partners</xsd_2:agent>
    </xsd_2:emetteur>
    <xsd_2:destinataire>
      <xsd_2:agent>MacroNet Corp.</xsd_2:agent>
    </xsd_2:destinataire>
  </xsd_2:enteteMessage>
  <xsd_2:evenementServeurActe>
    <xsd_2:patient>
      <xsd_2:identifiant>
        <xsd_2:numeroIdentifiantSante>
          <xsd_2:identifiant>925287F4-20A4-11F0-B7B4-EA3B43076670</xsd_2:identifiant>
        </xsd_2:numeroIdentifiantSante>
        <xsd_2:numeroIdentifiantPatients>
          <xsd_2:numeroIdentifiantPatient>
            <xsd_2:identifiant>925287F4-20A4-11F0-B7B4-EA3B43076670</xsd_2:identifiant>
            <xsd_2:autorite>
              <xsd_2:nom>http://hospital.smarthealthit.org</xsd_2:nom>
            </xsd_2:autorite>
          </xsd_2:numeroIdentifiantPatient>
        </xsd_2:numeroIdentifiantPatients>
      </xsd_2:identifiant>
      <xsd_2:personnePhysique>
        <xsd_2:nomNaissance>Chadwick</xsd_2:nomNaissance>
        <xsd_2:prenoms>
          <xsd_2:prenom>Zeke</xsd_2:prenom>
        </xsd_2:prenoms>
      </xsd_2:personnePhysique>
    </xsd_2:patient>
  </xsd_2:evenementServeurActe>
</xsd_2:evenementsServeurActes>
```

Visual Trace

-

Étape Processus

requêtant un serveur

FHIR sur la base des

identifiants fournis par

le document HPRIM

XML

Ex:

Patient?identfier=http://

/hospital.smarthealthit.

org|925287F4-20A4-

11F0-B7B4-

EA3B43076670

-

Trace de la réponse du

serveur FHIR

VISUAL TRACE

Session ID: 24108 Legend Printable Version Go to items 1 - 34 Items per page 200 Show events Show internal items Apply Filter None Previous Page Next Page Previous Session Next Session

IO	Services	Processes	Operations	IO
Service IO	source HPRIM XML 2 4	HPRIM XML	FHIR GET / irisapp/fhir/r4	target HPRIMXML 2 4
Operation IO				Operation IO

[1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23] [24] [25]

Time

Time

Time

Header **Body** **Contents**

ID: 7529886

Type: Trace

Text: FHIR Response: {"resourceType":"Bundle","id":"f8f9c7ed20db-11f0-b7b4-ea3b43076670","type":"searchset","timestamp":"2025-04-24T07:15:59Z","total":1,"link":[{"relation":"self","url":"http://irishealth:52773/irisapp/fhir/r4/Patient?identfier=http%3A%2F%2Fhospital.smarthealthit.org%7C925287F4-20A4-11F0-B7B4-EA3B43076670"}],"entry":[{"fullUrl":"http://irishealth:52773/irisapp/fhir/r4/Patient/568894","resource":{"resourceType":"Patient","text":{"status":"generated","div":"<div xmlns='\"http://www.w3.org/1999/xhtml\"'>Generated by Synthea.Version identifier: 65d0325\\n . Person seed: 2346613998840497628 Population seed: 1596649671047</div>"},"extension":[{"url":"http://hl7.org/fhir/us/core/StructureDefinition/us-core-race","extension":{"url":"ombCategory","valueCoding":{"system":"urn:oid:2.16.840.1.113883.6.238","code":"2106-3","display":"White"}},{"url":"http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity","extension":{"url":"ombCategory","valueCoding":{"system":"urn:oid:2.16.840.1.113883.6.238","code":"2186-5","display":"Not Hispanic or Latino"}},{"url":"text","valueString":"Not Hispanic or Latino"}]}],"url":"http://hl7.org/fhir/StructureDefinition/patient-mothersMaidenName","valueString":"Delfina519 Dietrich576"},"url":"http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex","valueCode":"M"},"url":"http://hl7.org/StructureDefinition/patient-birthPlace","valueAddress":{"city":"Rowley","state":"Massachusetts","country":"US"},"url":"http://synthetichealth.github.io/synthea/disability-adjusted-life-years","valueDecimal":0.0},"url":"http://synthetichealth.github.io/synthea/quality-adjusted-life-years","valueDecimal":44.0},"identifier":{"system":"https://github.com/synthetichealth/synthea","value":"a5b737fb-1f94-4767-802b-1112051f5101"},"type":{"coding":{"system":"http://terminology.hl7.org/CodeSystem/v2-0203","code":"MR","display":"Medical Record Number"}},"text":"Medical Record Number"},"system":"http://hospital.smarthealthit.org","value":"925287F4-20A4-11F0-B7B4-EA3B43076670"},"type":{"coding":{"system":"http://terminology.hl7.org/CodeSystem/v2-0203","code":"SS","display":"Social Security Number"},"system":"http://hl7.org/fhir/sid/us-ssn","value":"999-16-6498"},"type":{"coding":{"system":"http://terminology.hl7.org/CodeSystem/v2-0203","code":"DL","display":"Driver's License"},"text":"Driver's License"},"system":"urn:oid:2.16.840.1.113883.4.3.25","value":"S99950276"},"type":{"coding":{"system":"http://terminology.hl7.org/CodeSystem/v2-0203","code":"PPN","display":"Passport Number"},"text":"Passport Number"},"system":"http://standardhealthrecord.org/fhir/StructureDefinition/passportNumber","value":"X39628570X"},"name":{"use":"official","family":"Hendrix","given":["Jimi"],"prefix":["Mr."]},"telecom":{"system":"phone","value":"555-769-8353","use":"home"},"gender":"male","birthDate":"1942-11-27","address":{"extension":{"url":"http://hl7.org/fhir/StructureDefinition/geolocation","extension":{"url":"latitude","valueDecimal":41.777141058411324},"url":"longitude","valueDecimal":-71.13123849523227}}},"line":{"360 Lueftgen Run"},"city":"Somerset","state":"Massachusetts","postalCode":"02725","country":"US"},"maritalStatus":{"coding":{"system":"http://terminology.hl7.org/CodeSystem/v3-MaritalStatus","code":"M","display":"M"},"text":"M"},"multipleBirthBoolean":false,"communication":{"language":{"coding":{"system":"urn:ietf:bcp:47","code":"en-US","display":"English"},"text":"English"},"meta":{"lastUpdated":"2025-04-24T00:41:00Z","versionId":"1"},"id":"568894"},"search":{"mode":"match"}}}]}

Logged: 2025-04-24 09:15:59.554

Source: FHIR GET / irisapp/fhir/r4

Session: 24108

Job: 6219

Class: FHIR.client

Method: get

Trace: user

Visual Trace

- Étape Opération envoyant le document HPRIM XML enrichi avec les données <personnePhysique> provenant du serveur FHIR

- Trace du document HPRIM XML envoyé à la cible

VISUAL TRACE

Session ID: 24108 Legend Printable Version Go to items 1 - 34 Items per page 200 Show events Show internal items Apply Filter None Previous Page Next Page Previous Session Next Session

IO	Services	Processes	Operations	target HPRIMXML 2	IO
Service IO	source HPRIM XML 2	HPRIM XML	FHIR GET / irisapp/fhir/r4	4	Operation IO

[1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23] [24] [25]

Time →

Time →

Header Body Contents

View Full Contents View Format Raw

Expand All

```
<?xml version="1.0" ?>
<!-- type: EnsLib.EDI.XML.Document id: 65 -->
<xsd_2:evenementsServeurActes xmlns:xsd_3="http://www.hprim.org/inseeXML"
xmlns:xsd_2="http://www.hprim.org/hprimXML">
  <xsd_2:enteteMessage>
    <xsd_2:dateHeureProduction>2025-04-24 01:22:44</xsd_2:dateHeureProduction>
    <xsd_2:emetteur>
      <xsd_2:agents>
        <xsd_2:agent>GigaComp Partners</xsd_2:agent>
      </xsd_2:agents>
    </xsd_2:emetteur>
    <xsd_2:destinataire>
      <xsd_2:agents>
        <xsd_2:agent>MacroNet Corp.</xsd_2:agent>
      </xsd_2:agents>
    </xsd_2:destinataire>
  </xsd_2:enteteMessage>
  <xsd_2:evenementServeurActe>
    <xsd_2:patient>
      <xsd_2:identifiant>
        <xsd_2:numeroIdentifiantSante>
          <xsd_2:identifiant>925287F4-20A4-11F0-B7B4-EA3B43076670</xsd_2:identifiant>
        </xsd_2:numeroIdentifiantSante>
        <xsd_2:numeroIdentifiantPatients>
          <xsd_2:numeroIdentifiantPatient>
            <xsd_2:identifiant>925287F4-20A4-11F0-B7B4-EA3B43076670</xsd_2:identifiant>
            <xsd_2:autorite>
              <xsd_2:nom>http://hospital.smarthealthit.org</xsd_2:nom>
            </xsd_2:autorite>
          </xsd_2:numeroIdentifiantPatient>
        </xsd_2:numeroIdentifiantPatients>
      </xsd_2:identifiant>
      <xsd_2:personnePhysique>
        <xsd_2:nomNaissance>Hendrix</xsd_2:nomNaissance>
        <xsd_2:prenoms>
          <xsd_2:prenom>Jimi</xsd_2:prenom>
        </xsd_2:prenoms>
        <xsd_2:dateNaissance>
          <xsd_2:date>1942-11-27</xsd_2:date>
        </xsd_2:dateNaissance>
      </xsd_2:personnePhysique>
    </xsd_2:patient>
  </xsd_2:evenementServeurActe>
</xsd_2:evenementsServeurActes>
```

Processus
FHIRContentAugmentation



Management Portal

Server IRISHealth Namespace IRISAPP User SuperUser Licensed To InterSystems IRIS Community Instance IRIS

Interoperability > Business Process Designer - (HPRIMXML.process.FHIRContentAugmentation)

New Open Save Save As Compile

75%

-Add Activity-

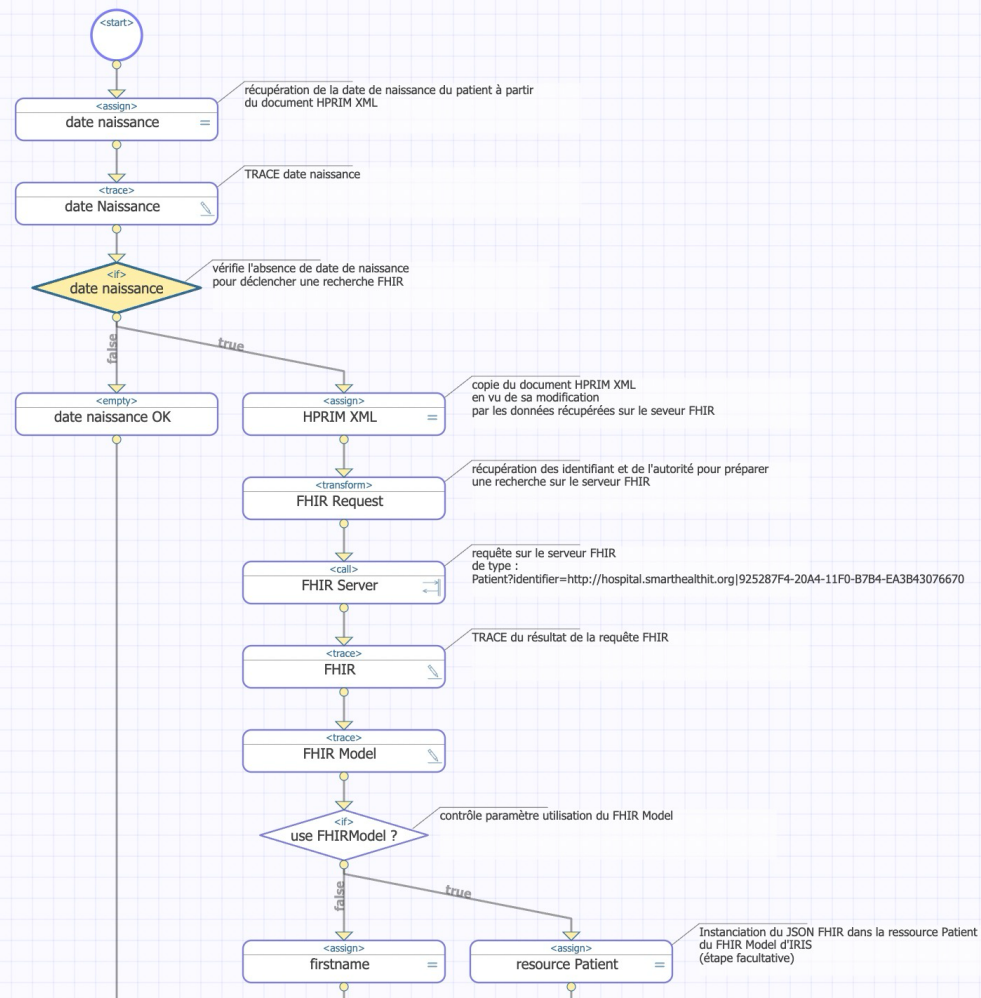
-Group Items-



Business Process

HPRIMXML.process.FHIRContentAugmentatic

Last modified: Today, 06:24:47AM



Transformation
FHIRRequest
récupérant les
identifiants pour
préparer la requête
FHIR sur la ressource
Patient:
/Patient?identifiant=...

DTL EDITOR

HPRIMXML.transfo.FHIRRequest

EnsLib.EDI.XML.Document
msgEvenementsServeurActes2_4:evenementsServeurActes

Create new SET

FHIR.request

New Open Save Save As Compile Test

source

enteteMessage

choice

choice.1 evenementServeurIntervention

choice.2 evenementServeurActe

choice.2.1 dateAction

choice.2.2 acteur

choice.2.3 bonCommande

choice.2.4 patient

choice.2.4.1 identifiant

choice.2.4.1.1 emetteur

choice.2.4.1.2 recepteur

choice.2.4.1.3 numeroidentifiantSante

choice.2.4.1.4 numeroidentifiantPatients()

choice.2.4.1.4.1 identifiant

choice.2.4.1.4.2 autorite

choice.2.4.1.4.2.1 nom

choice.2.4.1.4.2.2 OID

choice.2.4.1.4.2.3 type

choice.2.4.1.4.3 dateDebutValidite

choice.2.4.1.4.4 dateFinValidite

choice.2.4.2 personnePhysique

target

RequestPath

QueryString

Actions (0 selected)

Select all

✕ ✂ 📄 📁 🗑️ ⬆️ ⬇️ ⚙️ Disable New

☐ 1 SET autorite = source.{choice.evenementServeurActe.patient.identifiant.numeroidentifiantPatients(1).autorite.nom}

☐ 2 SET identifiant = source.{choice.evenementServeurActe.patient.identifiant.numeroidentifiantPatients(1).identifiant}

☐ 3 SET target.RequestPath = "/Patient"

☐ 4 SET target.QueryString = "identifiant="_autorite_"_"_identifiant

Transform Settings

DTL Name
HPRIMXML.transfo.FHIRRequest

Description

Source Class*
EnsLib.EDI.XML.Document

Source Document Type
msgEvenementsServeurActes2_4:evenementsServeurActes

Target Class*
FHIR.request

Target Document Type

Mode
Create new

☒ Report errors

☒ Ignore missing source segments and properties

☐ Treat empty repeating fields as null

☐ Allow empty segments in target

Language
ObjectScript

Python From/Import statements

Classe [FHIR.client](#)
requêtant un serveur
FHIR

```
Class FHIR.client Extends Ens.BusinessOperation
{

Property FHIREndpoint As %String [ InitialExpression = "/irisapp/fhir/r4" ];

Parameter SETTINGS = "FHIREndpoint:FHIR";

Method get(pRequest As FHIR.request, Output pResponse As FHIR.response) As %Library.Status
{
    try{
        set sc = $$$OK
        set pResponse = ##class(FHIR.response).%New()
        set FHIRInstance = ##class(HS.FHIRServer.RestClient.FHIRService).CreateInstance(..FHIREndpoint)
        set FHIRRequest = FHIRInstance.MakeRequest("GET",,,pRequest.RequestPath,pRequest.QueryString)
        set FHIRResponse = FHIRInstance.InvokeRequest(FHIRRequest)
        set pResponse.json = FHIRResponse.Json
        $$$TRACE("FHIR Response: " _ FHIRResponse.Json.%ToJSON())
    } catch exp{
        set sc = exp.AsStatus()
    }
    return sc
}

XData MessageMap
{
    <MapItems>
        <MapItem MessageType="FHIR.request">
            <Method>get</Method>
        </MapItem>
    </MapItems>
}
```